

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CE) Examination

May 2014

CE207 Java Programming

Date: 03.05.2014, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q-1 (a) Which is the default layout for an applet? [01]
(b) Explain how exception handling mechanism can be used in a program? [04]
(c) Differentiate the following file handling classes [06]
1) OutputStream and InputStream
2) Reader and Writer

- Q-2 (a) What are the differences between an Applet and an Application? [02]
(b) Explain the following terms. [04]
1) paint
2) repaint
3) update
4) font
(c) Draw and explain life cycle of an Applet. [06]

OR

- Q-2 (a) Why do we need finally? Justify. [02]
(b) Give code for which two catch blocks are triggered by the same exception. Prove that only the first catch block is executed. [04]
(c) Draw and explain life cycle of a Thread. [06]

- Q-3 (a) Write a program to create a thread by implementing runnable interface and by inhering thread class. [04]
(b) Write a program to read primitive data types from a file. [04]
(c) What is deadlock? How to remove deadlock situation in multithreading. [04]

OR

- Q-3 (a) What is an exception? Write a program to demonstrate the use of try catch block to catch the different types of exceptions. [04]
(b) What are the thread priorities available in thread class? Write a program to create 3 threads t1, t2 and t3 and set thread t1's priority to 2, t2's priority to 10 and t3's priority to 7. [04]
(c) What is Wrapper class? What is the use of it and write a program to convert float data type value to object. [04]

SECTION – II

- Q-4 (a) Justify the statement. "Java is platform independent". [01]
(b) What are the different types of operators available in java? [04]
(c) List out features of object oriented programming system. And explain any two. [06]

- Q-5 (a) Is it legal for a static method to invoke a non-static method? Justify. [02]
(b) What is the default length and capacity of stringbuffer constructor? And write a program to compare two string data type variables that prints 1 if both matches otherwise prints 0. [04]
(c) Write a java program that declares a class box having three instance variables w, h and d as width, height and depth respectively. Initialize all the instance variables using constructor. And create one method to calculate volume of the box. [06]

OR

- Q-5 (a) What message does the compiler may give if we try to compile a program containing the statements:
 S="hello";
 String s; [02]
(b) What is an object and class? Explain with an example. [04]
(c) Write a java program which initialize array of 10 integer elements. Find out minimum and maximum element out of 10 elements. Also calculate average of 10 elements. [06]

- Q-6 (a) What is the difference between overriding and overloading a method? Explain with an example. [06]
(b) What is the use of inheritance? Explain types of inheritance? How java achieves multiple inheritance? [06]

OR

- Q-6 (a) Differentiate the following [06]
 1) interface and class
 2) static variable and instance variable
(b) Write a program in java. A class teacher contains two fields: name and qualification. Extends the class to department . It contains: dept_no and dept_name. And interface name college contains one field college_name . Using the above classes and interface get the appropriate information and display it. [06]
